



AI-Powered Learning Assistant with a RAG-Based Doubt Solver

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Abstract

- An AI-powered learning assistant where students upload course PDFs and ask doubts in plain English. A RAG pipeline (embeddings + Pinecone semantic search + Groq LLM) on the MERN stack grounds each answer in the student's own material.
- Outcome:** trustworthy answers with document and page citations.

Problem Statement & Objectives

- Problem:** keyword search misses rephrased questions, and generic chatbots give confident answers not grounded in course content.
- Objective 1:** Let students upload PDFs and ask natural-language questions.
- Objective 2:** Retrieve relevant passages using semantic (vector) search.
- Objective 3:** Generate LLM answers grounded in sources, with citations.
- Scope:** students querying their own course documents (PDFs).

Existing vs Proposed System

Feature	Existing System	Proposed System
Search method	Keyword-based matching	Semantic vector search
Answer source	General chatbot knowledge	Student's own uploaded PDFs
Citations	None / not traceable	Document & page citations
Rephrased queries	Often missed	Matched by meaning
Limitation / Gap	Ungrounded, risk of misinformation	RAG ties answers to sources

Feasibility Study

Technical: Built using React.js, Node.js, Express.js, MongoDB, Pinecone, embeddings, and Groq LLM APIs; all components are readily available and integrate effectively.

Economic: Uses mostly open-source technologies and free/low-cost API tiers, making implementation affordable for student projects.

Operational: Simple PDF upload-and-query workflow allows students to interact with their course material without specialized technical knowledge.

Results & Testing

- Successfully implemented the RAG-based learning assistant.
- Semantic retrieval accurately identifies relevant course-content chunks.
- Groq LLM generates context-aware answers from retrieved material.

Conclusion

- working RAG-based learning assistant for course-specific doubt resolution.
- Combines semantic search, document retrieval, and LLMs to provide context-aware answers.
- Helps students quickly understand course material while maintaining accuracy.

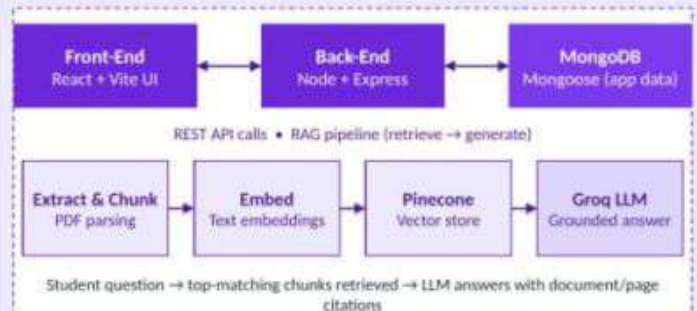
Future Scope

- Support more formats (DOCX, slides) beyond PDFs.
- Add personalized learning and quiz-generation features.
- Enable multi-language support for broader accessibility.

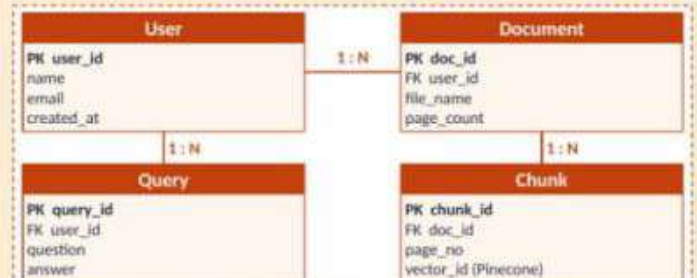
Tools & Technologies

- Front-End:** React.js, Vite, JavaScript, HTML/CSS
- Back-End:** Node.js, Express.js + REST APIs
- Database:** MongoDB + Mongoose (app data); Pinecone (vector DB)
- AI / RAG:** Text embeddings; Groq API (LLM)

System Architecture



ER Diagram & Database Design



Implementation Screenshots

